number of infinitesimal time intervals, *the amplitude is an integral over all possible paths between the end points*. The elegant fact that Feynman discovered is that the amplitude for each path bears a simple relation to a familiar expression from classical mechanics—the action for that path. The exact expression for each path is

$$e^{iA/\hbar},$$

where *A* is the action for the individual path.

Feynman's formulation can be summarized by a single equation:

$$C_{1,2} = \int_{paths} e^{iA/\hbar}. \qquad (9.26)$$

The path integral formulation is not merely an elegant mathematical trick; it has real power. In fact, it can be used to derive both Schrödinger equations, and all the commutation relations of quantum mechanics. But it really comes into its own in the context of quantum field theory, where it is the principal tool for formulating the laws of elementary particle physics.